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mechanisation. It is the use of control systems (such as numerical control, programmable logic control and other industrial control systems), along with other applications of information technology (such as computer-aided design and computer-aided manufacturing), to control industrial machinery and processes, thereby reducing the need for human intervention.

Different types of automation tools exist: Artificial neural network, distributed control system, human-machine interface, SCADA, PLC, programmable automation controller, instrumentation, motion control and robotics.

A simple process control loop consists of a measurement system, a controller and a final control element.

Keep abreast of emerging trends

To get an edge over your competitors, learn advanced process control with respect to the underlying theory, implementation studies, the benefits that its applications will bring and projections of future trends.

In particular, the following trends are redefining automation:

Industrial wireless. Wireless connectivity is lowering the consumption of copper cabling, and associated costs of laying cable trenches and securing the safety of networks that cut across hazardous or difficult-to-reach zones. So automation systems can cover more sensory points and relay important information to moving personnel, bringing down the potential response times.

Embedded intelligence and machine-to-machine (M2M) communications. With increasing memory and computing power, embedded intelligence will grow. And so will M2M and M2H (machine-to-human) communications. These will force adoption of interface standards like OMAC. Understanding these nuances is important to redefine productivity and flexibility norms in future factories. With better understanding, engineers will also be able to integrate energy management and

environment control strategies with different kinds of recipe management or production cycles.

Web services and applications integration. Maintenance and audit functions can now be conducted without specialists having to fly down to the machine location. They can connect through Web services. If a human intervention (change of parts or in the application software) is needed, the local person can be guided to make the change.

Real-time information to boost productivity. Real-time access to information is an essential feature of industrial automation systems as these are meant to execute control commands to the process. Recent developments in the area of 'real-time' production control have been captured in the shape of manufacturing execution systems. This is a great management tool in the hands of operational personnel.

Robots. Lighter and smarter robots are dominant in fast-moving consumer goods assembly and specialised packaging. Within the core manufacturing such as automotive, the trend of heavy-duty robots is seen, with designs for payloads lower than 100 kg. These require about 25 per cent less space, while being consistently more robust and having significantly lower maintenance requirements.

Looking ahead

There are bound to be implications of the global turmoil in India as well but on a mid-term and long-term horizon, things do look good. After all, we are a big country with a lot of potential for automation for want of improvement in quality, manufacturing time or safety. So it is quite encouraging to work in this segment. The concept of a no-man factory is far-fetched; there could be work cells or process zones that are well automated and unmanned, but even these would need human surveillance or intervention for maintenance as mechanical parts are prone to long-term wear. ●

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